Participation E 2

For special participation part E, I asked DeepSeek to teach me how to improve a factor's performance in quantitative investing: https://chat.deepseek.com/share/hicc0bcs9s3x02hm5x

Summary of Achievements

Through our iterative collaboration, I have made significant progress in quantitative factor development and refinement. Here are the key achievements:

1. In-Depth Problem Identification

- Successfully identified a specific performance anomaly: **abnormal returns in the tail portion (small factor values)** of my originally constructed factor.
- Conducted systematic factor evaluation, pinpointing that the core issue lies in **factor construction logic** rather than superficial parameter tuning.

2. Structured Analysis & Hypothesis Generation

- Developed a structured analysis framework to dissect the factor's underlying logic, identifying multiple potential root causes:
  - **Weight distortion in extreme markets** due to the `clip()` function
  - **Standardization failure in tail regions** during high volatility
  - **Signal misalignment** in price-volume divergence scenarios
  - **Multi-cycle signal conflict** in tail market conditions

3. Systematic Improvement Methodology

- Implemented a **methodical testing approach** by developing independent, modular improvement versions for isolated testing
- Created multiple specialized solutions targeting tail market behavior:
  - **Dynamic weighting mechanisms** (Sigmoid transformation)
  - **Tail-adaptive standardization** with volatility adjustment
  - **Divergence-aware signal processing**
  - **Market-state-based factor fusion**
  - **Microstructure-informed enhancements** (order flow imbalance, liquidity risk, market entropy)

4. Data Utilization Expansion

- Expanded beyond basic OHLCV data to incorporate **additional available fields**:
  - `taker_buy_turnover` for order flow analysis
  - `trade_count` for trade intensity metrics
  - `high-low ranges` for price efficiency calculations
  - **Comprehensive microstructure indicators** from available Binance data

5. Innovative Factor Construction Approaches

- Explored several novel quantitative finance concepts:
  - **Order Flow Imbalance (OFI)** enhanced factors
  - **Liquidity black hole detection** algorithms
  - **Multi-scale sentiment resonance** indicators
  - **Market microstructure entropy** measurements
  - **Pressure index** formulations for extreme markets

6. Rigorous Testing Discipline

- Maintained **strict avoidance of lookahead bias** in all implementations
- Established clear **version control and isolation** for testing different improvement hypotheses
- Developed **tail-specific evaluation focus** to directly address the identified performance issue

7. Refined Problem Understanding

- Progressed from recognizing a performance anomaly to understanding that **tail market behavior requires fundamentally different treatment**
- Recognized that **cryptocurrency markets exhibit unique microstructure characteristics** in extreme conditions
- Developed insights into how **different market regimes** (normal vs. tail) require distinct factor logic

Overall Achievement:

I have transformed from having a factor with known performance issues to possessing:

1. A **systematic framework** for factor diagnosis and improvement
2. Multiple **tested hypotheses** about tail market behavior
3. **Specialized tools** for cryptocurrency market microstructure analysis

4. A **methodology** for creating regime-aware quantitative factors

While the specific factor performance improvements may still require further refinement, I have established a robust process for quantitative research that will yield dividends in future factor development projects. The most valuable outcome is the development of my analytical framework and testing methodology, which transcends any single factor implementation.